

Lesson 2: Everything Is a Widget

FlutterLearn | Module 1: Flutter Foundations

The one idea that matters

In Flutter, everything is a widget. Text is a widget. A button is a widget. Padding, alignment, rows, columns — widgets. Even your whole app is a widget.

A widget is just a description of what the UI should look like. You don't create buttons and move pixels around; you describe the UI you want, and Flutter figures out how to draw it. When something needs to change, you describe the new UI, and Flutter updates the screen.

Widgets form a tree: big widgets contain smaller widgets, which contain even smaller ones. That nesting you saw in Lesson 1 *is* the widget tree.

StatelessWidget

The simplest kind of widget is a StatelessWidget — a widget whose appearance never changes once it's built. A label, an icon, a static title: all stateless.

Every stateless widget overrides one method: `build()`. It takes a `BuildContext` and returns the widget tree for this widget:

```
class Greeting extends StatelessWidget {
  final String name;

  const Greeting({super.key, required this.name});

  @override
  Widget build(BuildContext context) {
    return Text(
      'Hello, $name!',
      style: const TextStyle(fontSize: 24),
    );
  }
}
```

Notice three things: the data comes in through the constructor (`name`), the widget is immutable (`final`), and `build()` simply returns what should appear. No step-by-step drawing instructions anywhere.

You'd use it like any other widget: `Greeting(name: 'Aarav')`.

Widgets you will use every day

- Text — displays a string, with optional styling.
- Icon — a Material icon, e.g. Icon(Icons.star).
- Image — shows an image from assets or the network.
- ElevatedButton — a raised button with an onPressed callback.

```
ElevatedButton(  
  onPressed: () {  
    print('Button tapped!');  
  },  
  child: const Text('Tap me'),  
)
```

Stateless vs Stateful — a preview

StatelessWidget can't change after it's built. But apps need counters, forms, toggles — things that change. That's what StatefulWidget is for, and it's the entire subject of Lesson 4.

Key terms

- Widget tree — the nested structure of widgets that describes your UI.
- build() — the method that returns a widget's UI description.
- Immutable — once created, a stateless widget's properties can't change.